

# VIII. Mini Project - Water Consumption Analysis

## AIM:

To clean, process, and visualize water consumption data using Python, Power BI, and Tableau, and to build clear visualizations for identifying consumption trends, city-wise variations, and user consumption patterns.

## DOMAIN:

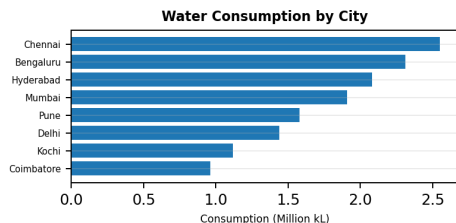
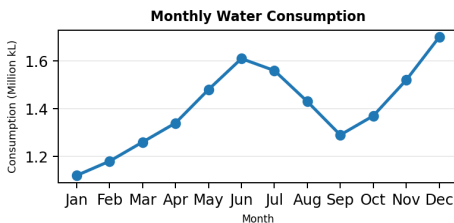
Water Consumption Analytics

## PROBLEM STATEMENT:

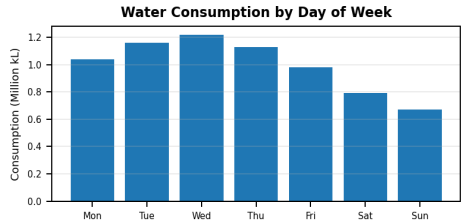
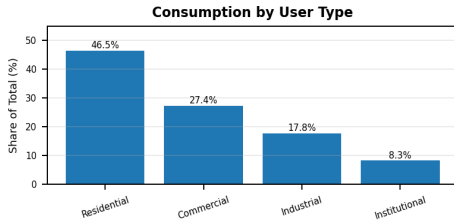
With increasing water usage across different cities and consumer categories, it is important to understand how water consumption changes over time and across different areas. There is a need for a data-driven visualization system to compare water usage by month, city, and user type and present the results through interactive dashboards for easier analysis.

## OUTPUT:

### Python:



*Monthly water consumption and city-wise consumption visualizations generated using Python.*



*User-type and day-of-week water consumption visualizations generated using Python.*

## PowerBI:



*Power BI dashboard for Water Consumption Analysis.*

Tableau:

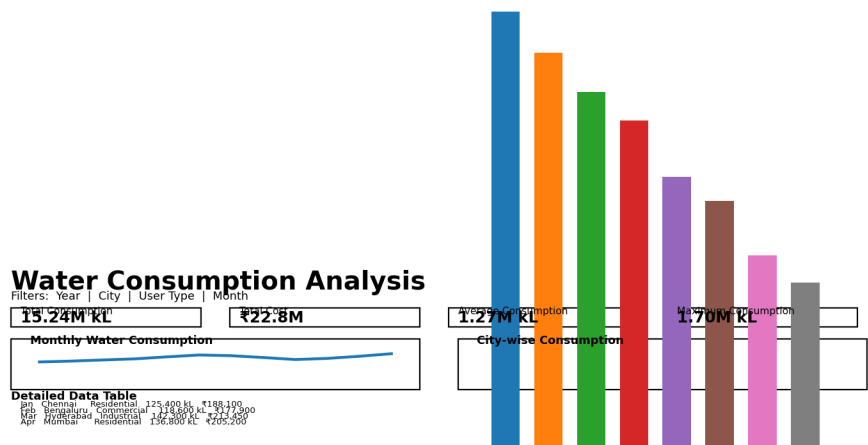


Tableau dashboard for Water Consumption Analysis.

RESULT:

Thus, the water consumption data was successfully analyzed and visualized using Python, Power BI, and Tableau. The visualizations and dashboards provide a clear view of consumption trends, city-wise differences, user consumption patterns, and detailed water consumption information.